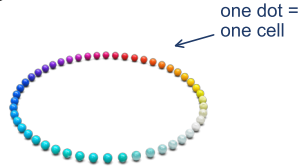


a

Same hole, different geometry

b

Cells trace a state-space loop

c

How long the loop survives
 = this paper's $\max H_1$ persistence

